

Beyond Semantic Confidence: Relation-Algebra-Constrained Geometric Compatibility for 3D Scene Graph Relations

Supplementary Material

Anonymous submission

Model and Preprocessing Details

Open3DSG. Open3DSG uses the official non-averaged BLIP checkpoint `epoch=13-step=13104.ckpt`, selected by train/dev validation loss. The preprocessing recovery covers all 548 evaluation contexts by setting the minimum visible-object count to two and regenerating relaxed views for two scans. The unmodified public-source route covers 533/548 contexts. The release manifest records the checkpoint hash, preprocessing configuration, and recovered-context inventory.

SGFN. We use the official `full_1160` source with its 160-object, 26-relation checkpoint. The evaluated scans, denominator, methods, and tensor-shape checks are recorded in the supplemental package.

Compatibility and Structural Diagnostics

RelCompat3D fits one logistic model per measured relation family. Numeric features use training-only means and standard deviations; missing values use the corresponding training mean. Inputs are predicate/family indicators T , predicate-independent pair geometry G , and explicit $T \times G$ interactions. Source score, source rank, source identity, and object class are forbidden. Orbit-consistent augmentation preserves the proximity predicate under endpoint swap; vertical order swaps endpoints and applies the inverse predicate. Linked counterfactuals receive the fixed margin loss in the main paper. Support/contact receives the pairwise loss but no blanket endpoint transform.

Table 1: **Training and development discrimination.** Labels comprise GT-positive/counterfactual pairs, not human physical-validity judgments.

Family	Train	Dev	Param.	Train pairs	AUROC	Brier
Proximity	20,902	2,106	22	9,562	1.0000	.0009
Vertical order	4,922	566	23	1,644	.9982	.0079
Support/contact	34,384	3,574	24	22,755	.9933	.0256

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Across all 6,246 development rows, AUROC/AUPRC/Brier are .9973/.9973/.0157. On 3,516 linked development pairs, the positive wins 99.23% of the time with mean logit margin 14.13. Correct-predicate compatibility wins on 97.40% of 384 vertical positives. Exact orbit projection gives maximum proximity-swap and vertical-inverse errors of zero on development and all three final source outputs.

The factor-isolation ablation uses the same split and estimator class. At development, T -only, true raw- G -only, additive $T + G$, and interaction $T \times G$ obtain AUROC .6124/.8809/.9193/.9822 and Brier .2370/.1363/.1061/.0495. Shuffled and mismatched-pair geometry reduce the intended reliability signal. These results establish factor use and exact relation algebra; they do not convert the constructed target into independent physical-validity ground truth.

Scan-Cluster Bootstrap Sensitivity

The main analysis resamples 548 relation contexts. To test dependence among contexts from the same scene, we additionally resample the 157 scan identifiers 1,000 times and carry every context of a sampled scan together. The ranking, model, and point estimates remain unchanged; shared bootstrap indices are used across conditions.

Table 2: **Scan-cluster sensitivity at $K = 100$.** Entries are RelCompat3D product minus source score with paired 95% intervals.

Source	Δ Recall	Δ Verifier-V
VL-SAT	+0.053 [.0000,.0119]	−.0151 [−.0171,−.0135]
Open3DSG	+0.0894 [.0646,.1119]	−.0903 [−.0959,−.0846]
SGFN	+0.0184 [.0137,.0230]	−.0258 [−.0285,−.0233]

All Verifier-V intervals remain below zero. Recall intervals exclude zero for Open3DSG and SGFN; the VL-SAT lower bound reaches zero, consistent with its near-saturated Recall@100 baseline.

Family Composition at the Global $K = 100$ Operating Point

Table 3 reports each family’s slice inside the actual global top-100 ranking. Deltas compare the RelCompat3D product

with the source score using shared paired context-bootstrap indices.

Table 3: **Global-top-100 family slices.** Positive ΔV is a regression. Values are verifier-derived.

Source	Family	ΔR	ΔV
VL-SAT	Support/contact	+0.0028	+0.0076
VL-SAT	Proximity	+0.0079	-.0078
VL-SAT	Vertical order	+0.0051	-.0586
Open3DSG	Support/contact	-.0341	+0.0036
Open3DSG	Proximity	+0.1959	-.0457
Open3DSG	Vertical order	+0.1821	-.2894
SGFN	Support/contact	+0.0017	+0.0077
SGFN	Proximity	+0.0368	-.0152
SGFN	Vertical order	+0.0128	-.0617

Support/contact V regresses in all three source rankings, while vertical order drives most of the aggregate reduction. The result is consequently a scoped operating-point improvement, not a uniform family claim. Full within-family and global-slice results for $K \in \{5, 10, 20, 50, 100\}$, including simultaneous family-wise intervals, are included in the supplemental package.

Relative-Size Scope Extension

We evaluate bigger than/smaller than as secondary breadth evidence, not as core evidence that learned compatibility is necessary. The extension uses the same 1,061/117/157 train/development/evaluation split. Its model receives predicate sign, robust point extents and volume ratios, and signed predicate-geometry interactions; source score, rank, object class, and evaluation labels are excluded. Alternating point subset A (p05-p95 extents) provides model evidence, whereas disjoint subset B (p02-p98 extents) provides the verifier measurement. Annotation OBB volume is used only by a fixed-rule baseline.

The model fits 3,244 training rows and obtains target-discrimination AUROC .9974 on 376 development rows. Correct-predicate ordering is 96.81%; exact inverse-equivariance and common-scale invariance have maximum error 2.22×10^{-16} . Table 4 summarizes the $K = 100$ result; the supplemental package contains all five K values and family compositions.

Table 4: **Relative-size extension at $K = 100$.** Within-size Recall is unchanged for all sources. Global columns use the four-family denominator of 4,142; parentheses are paired 95% context-bootstrap intervals.

Source	Within-size ΔV	Global ΔR	Global ΔV
VL-SAT	-.1025 [-.1063, -.0988]	+0.0048 [.0020, .0080]	-.0239 [-.0261, -.0218]
Open3DSG	-.1274 [-.1383, -.1167]	+0.0698 [.0478, .0935]	-.1040 [-.1090, -.0997]
SGFN	-.0593 [-.0616, -.0572]	+0.0256 [.0199, .0317]	-.0359 [-.0381, -.0336]

The fixed robust-point rule is as strong or stronger than the learned model on within-family violation: learned-minus-rule ΔV is +.00318 for VL-SAT, +.00005 for Open3DSG,

and +.00367 for SGFN. Disjoint point subsets remove exact rule reuse but not shared-geometry construct dependence. The family therefore broadens the measured scope while remaining a rule-easy secondary analysis.

Verifier-Uncertainty Sensitivity

The primary $V@K_{all}$ includes uncertain rows in its denominator but does not label them satisfied. We therefore recompute each ranking with $V@K_{dec}$ over satisfied/violated rows, uncertainty rate $U@K$, and a pessimistic bound $V@K_{pess}$ that counts every uncertain row as a violation. Table 5 gives the primary $K = 100$ comparison.

Table 5: **Uncertainty sensitivity at $K = 100$.** Lower is better for all four reported quantities.

Source	Condition	V_{all}	V_{dec}	U	V_{pess}
VL-SAT	Source score	.0476	.0729	.3464	.3941
VL-SAT	RelCompat3D product	.0325	.0473	.3123	.3448
Open3DSG	Source score	.1242	.2128	.4164	.5406
Open3DSG	RelCompat3D product	.0339	.0465	.2714	.3052
SGFN	Source score	.0630	.1005	.3732	.4362
SGFN	RelCompat3D product	.0372	.0563	.3393	.3765

Algebra-product minus source-score changes in V_{dec} are -.0256 [-.0282, -.0229] for VL-SAT, -.1663 [-.1736, -.1594] for Open3DSG, and -.0442 [-.0472, -.0409] for SGFN. Uncertainty-rate changes are -.0341, -.1450, and -.0340; pessimistic-V changes are -.0493, -.2354, and -.0597. All corresponding paired 95% intervals exclude zero. The hard-rule baseline has zero all-status V but retains uncertain rows and therefore has pessimistic V of .3712/.4722/.4042. Thus zero measured violation does not imply reliable or complete selection.